Comprehensive Supplemental Material for "Information Loss is Local: The Gravitational Refractive Index as Petz Recovery Fidelity"

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I. S1. THREE ROUTES TO $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$

We present three consistent physical interpretations of the gravitational entropy production $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$, showing that all three yield the same quantity from different physical settings. For the exponential metric, $-g_{00} = e^{-r_{\mathrm{s}}/r}$, all three give $\Sigma_{\mathrm{grav}} = r_{\mathrm{s}}/r$ exactly. For the Schwarzschild metric, $-g_{00} = 1 - r_{\mathrm{s}}/r$, all three give $\Sigma_{\mathrm{grav}} = -\ln(1 - r_{\mathrm{s}}/r) \approx r_{\mathrm{s}}/r + O((r_{\mathrm{s}}/r)^2)$.

All three frameworks take the metric g_{00} as input and produce the same information-theoretic quantity Σ_{grav}. This convergence is a consistency check confirming that the identification $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$ is robust across different physical settings, not three independent derivations of Σ_{grav} from first principles.

A. Route A: Modular flow (Dorau–Much extension)

This route derives Σ_{grav} from the algebraic quantum field theory (AQFT) framework using the Bisognano–Wichmann theorem and the entanglement first law.

Step 1: Bisognano–Wichmann theorem.—For a Rindler wedge in Minkowski spacetime, the modular operator Δ_Ω associated with the vacuum state $|\Omega\rangle$ and the wedge algebra $\mathcal{A}(W)$ generates Lorentz boosts [1]:

$$\Delta_\Omega^{is} = e^{-2\pi K s}\,, \tag{1}$$

where K is the boost generator and s is the modular flow parameter.

Step 2: Extension to Schwarzschild.—For a static observer at radius r in the Schwarzschild spacetime, the local Rindler approximation gives the effective boost parameter scaled by the surface gravity $\kappa = 1/(2r_{\rm s})$ and the local redshift factor. The modular Hamiltonian for the wedge associated with the exterior region ($r > r_{\rm s}$) of a Schwarzschild black hole is [2]

$$H_{\rm mod}(r) = \frac{2\pi}{\kappa\sqrt{1 - r_{\rm s}/r}}\, H_{\rm local}\,, \tag{2}$$

where $H_{\rm local}$ is the local energy and the factor $1/\sqrt{1 - r_{\rm s}/r}$ is the Tolman blueshift.

Step 3: Entanglement first law.—The first law of entanglement entropy states that for small perturbations $\delta\rho$ about the vacuum, the change in quantum relative entropy (QRE) equals the change in modular energy:

$$\delta D(\rho\|\sigma_\beta) = {\rm Tr}[\delta\rho\, H_{\rm mod}] = \frac{2\pi}{\kappa\sqrt{1 - r_{\rm s}/r}}\, \langle\delta E\rangle_{\rm local}\,. \tag{3}$$

Step 4: Energy redshift.—An excitation of local energy $\langle\delta E\rangle_r$ at radius r is observed at infinity as

$$\langle\delta E\rangle_\infty = \sqrt{1 - r_{\rm s}/r}\, \langle\delta E\rangle_r\,. \tag{4}$$

Step 5: Fractional QRE loss.—The QRE at radius r, using Eqs. (3) and (4), is

$$D_{\rm in}(r) = \frac{2\pi}{\kappa\sqrt{1 - r_{\rm s}/r}}\, \langle\delta E\rangle_r = \frac{2\pi}{\kappa(1 - r_{\rm s}/r)}\, \langle\delta E\rangle_\infty\,. \tag{5}$$

The QRE at infinity is

$$D_{\rm out}(\infty) = \frac{2\pi}{\kappa}\, \langle\delta E\rangle_\infty\,. \tag{6}$$

The fractional QRE loss is therefore

$$\frac{D_{\rm in} - D_{\rm out}}{D_{\rm in}} = 1 - \frac{D_{\rm out}}{D_{\rm in}} = 1 - (1 - r_{\rm s}/r) = \frac{r_{\rm s}}{r}\,. \tag{7}$$

This is exact for Schwarzschild. The absolute QRE decrease is

$$\Sigma^{\rm (a)}_{\rm grav}(r) = D_{\rm in} - D_{\rm out} = D_{\rm in}\cdot\frac{r_{\rm s}/r}{1} = \frac{D_{\rm out}\cdot(r_{\rm s}/r)}{1 - r_{\rm s}/r}\,, \tag{8}$$

which in the weak-field limit ($r_{\rm s}/r \ll 1$) gives $\Sigma^{\rm (a)}_{\rm grav} \approx (r_{\rm s}/r)\, D_{\rm out}$. The normalization-independent result is

$$\boxed{\Sigma_{\rm grav} = -\ln(1 - r_{\rm s}/r) = -\ln(-g^{\rm Schw}_{00})\,.} \tag{9}$$

For the exponential metric ($-g_{00} = e^{-r_{\rm s}/r}$), the same analysis with modified Tolman factor gives $\Sigma_{\rm grav} = r_{\rm s}/r$ exactly.

B. Route B: Gravitational Landauer principle (Herrera 2020)

This route uses the thermodynamic cost of information erasure in a gravitational field.

Step 1: Landauer principle at infinity.—At spatial infinity (flat space), the minimum energy cost to erase one bit of information at temperature T_∞ is [3]

$$W_\infty = k_\mathrm{B} T_\infty \ln 2\,. \tag{10}$$

Step 2: Tolman relation.—In thermal equilibrium in a static gravitational field, the local temperature satisfies the Tolman relation [4]:

$$T(r)\,\sqrt{-g_{00}(r)} = T_\infty\,. \tag{11}$$

Therefore

$$T(r) = \frac{T_\infty}{\sqrt{-g_{00}(r)}}\,. \tag{12}$$

Step 3: Landauer cost at radius r.—The minimum erasure cost at radius r is

$$W(r) = k_\mathrm{B} T(r) \ln 2 = \frac{k_\mathrm{B} T_\infty \ln 2}{\sqrt{-g_{00}(r)}} = \frac{W_\infty}{\sqrt{-g_{00}(r)}}\,. \tag{13}$$

The excess cost relative to infinity is

$$\Delta W = W(r) - W_\infty = W_\infty \left(\frac{1}{\sqrt{-g_{00}}} - 1 \right). \tag{14}$$

Step 4: Dimensionless entropy production.—Following Herrera [5], the dimensionless gravitational entropy production is twice the logarithm of the cost ratio (factor of 2 because both the forward and reverse Landauer costs contribute):

$$\Sigma_\mathrm{grav}^{(\mathrm{b})} = 2 \ln\left(\frac{W(r)}{W_\infty} \right) = 2 \ln\left(\frac{1}{\sqrt{-g_{00}}} \right) = -\ln(-g_{00})\,. \tag{15}$$

Result.—

$$\boxed{\Sigma_\mathrm{grav}^{(\mathrm{b})} = -\ln(-g_{00})\,.} \tag{16}$$

For the exponential metric: $\Sigma_\mathrm{grav} = r_\mathrm{s}/r$. For Schwarzschild: $\Sigma_\mathrm{grav} = -\ln(1 - r_\mathrm{s}/r) \approx r_\mathrm{s}/r + \frac{1}{2}(r_\mathrm{s}/r)^2 + \cdots$.

Physical interpretation.—The gravitational entropy production is the excess informational cost of performing a bit-erasure operation at radius r compared to infinity. Gravity makes information processing more expensive at lower radii because the local temperature is higher (Tolman relation), and erasing a bit against a hotter bath requires more energy.

C. Route C: Pure-loss bosonic channel

This route models the gravitational redshift as a quantum optical channel.

Step 1: Gravitational frequency shift.—A photon of frequency ω_r emitted at radius r is detected at infinity with frequency

$$\omega_\infty = \sqrt{-g_{00}(r)}\,\omega_r\,. \tag{17}$$

Step 2: Pure-loss bosonic channel.—The redshift can be modeled as a pure-loss bosonic channel [6] with transmissivity

$$\eta = \frac{\omega_\infty^2}{\omega_r^2} = -g_{00}(r)\,, \tag{18}$$

where the intensity (energy) convention uses $\eta \propto \omega^2$ (accounting for both the frequency shift and the time dilation of the emission rate). The Kraus representation of a single-mode pure-loss channel with transmissivity η is [7]

$$\mathcal{N}_\eta(\rho) = \sum_{k=0}^{\infty} E_k\,\rho\,E_k^\dagger\,, \quad E_k = \sqrt{\binom{n}{k}}\,\eta^{n/2}(1-\eta)^{k/2}\,|n-k\rangle\langle n|\,. \tag{19}$$

This is a valid CPTP map for $\eta \in [0,1]$.

Step 3: QRE decrease.—For a pure-loss bosonic channel with transmissivity η, the decrease in quantum relative entropy between coherent states $|\alpha\rangle$ and $|\beta\rangle$ is [7]

$$\begin{aligned}\Delta D &= D(|\alpha\rangle\|\,|\beta\rangle) - D(\mathcal{N}_\eta|\alpha\rangle\|\mathcal{N}_\eta|\beta\rangle)\\ &= |\alpha-\beta|^2 - \eta|\alpha-\beta|^2 = (1-\eta)|\alpha-\beta|^2\,.\end{aligned} \tag{20}$$

The normalized (per-photon) entropy production is

$$\Sigma_{\text{grav}}^{(\text{c})} = -\ln\eta = -\ln(-g_{00})\,. \tag{21}$$

Result.—

$$\boxed{\Sigma_{\text{grav}}^{(\text{c})} = -\ln(-g_{00})\,.} \tag{22}$$

Remark on the factor of 2.—In the amplitude convention $\sqrt{\eta_{\text{amp}}} = \omega_\infty/\omega_r = \sqrt{-g_{00}}$, the transmissivity is $\eta_{\text{amp}} = -g_{00}$ and the QRE decrease is $-\ln(\eta_{\text{amp}}) = -\ln(-g_{00})$. In the intensity convention $\eta_{\text{int}} = (-g_{00})^{1/2}$ (frequency shift only, without the time-dilation contribution), one would get $\Sigma = -\ln((-g_{00})^{1/2}) = -\frac{1}{2}\ln(-g_{00})$. The correct answer requires including both the frequency shift and the time dilation of the emission rate, giving $\eta = -g_{00}$ and $\Sigma = -\ln(-g_{00})$.

D. Convergence of the three routes

All three routes yield the same universal formula:

TABLE I. Summary of the three derivation routes.

Route	**Physics**	**Key quantity**	**Result**
A (modular flow)	AQFT + entanglement first law	Fractional QRE loss	$-\ln(-g_{00})$
B (Landauer)	Tolman + erasure cost	Excess erasure cost	$-\ln(-g_{00})$
C (bosonic channel)	Redshift as pure loss	Channel transmissivity	$-\ln(-g_{00})$

The convergence of three independent routes—algebraic (modular flow), thermodynamic (Landauer), and quantum-optical (channel model)—provides strong evidence that $\Sigma_{\rm grav} = -\ln(-g_{00})$ is the correct identification, even though no single route constitutes a rigorous proof from first principles.

II. S2. PPN EXPANSION OF THE EXPONENTIAL METRIC

We provide a complete parametrized post-Newtonian (PPN) analysis of the exponential metric, demonstrating that it passes all current solar-system tests.

A. PPN parameters

In isotropic coordinates, the exponential metric is

$$g_{00}^{\rm exp} = -e^{-r_{\rm s}/r} = -\left(1 - \frac{r_{\rm s}}{r} + \frac{1}{2}\frac{r_{\rm s}^2}{r^2} - \frac{1}{6}\frac{r_{\rm s}^3}{r^3} + \cdots\right), \tag{23}$$

$$g_{ij}^{\rm exp} = e^{+r_{\rm s}/r}\,\delta_{ij} = \left(1 + \frac{r_{\rm s}}{r} + \frac{1}{2}\frac{r_{\rm s}^2}{r^2} + \cdots\right)\delta_{ij}\,. \tag{24}$$

Writing $U = r_{\rm s}/(2r) = GM/(c^2 r)$, the standard PPN form is

$$g_{00} = -(1 - 2U + 2\beta U^2 + \cdots)\,, \tag{25}$$

$$g_{ij} = (1 + 2\gamma U + \cdots)\,\delta_{ij}\,. \tag{26}$$

Comparing Eq. (23) with Eq. (25):

$$-g_{00} = 1 - 2U + 2U^2 + \cdots \quad \Longrightarrow \quad \beta = 1\,. \tag{27}$$

Comparing Eq. (24) with Eq. (26):

$$g_{ij} = (1 + 2U + \cdots)\,\delta_{ij} \quad \Longrightarrow \quad \gamma = 1\,. \tag{28}$$

$$\boxed{\gamma = \beta = 1 \quad \text{(identical to GR at 1PN).}} \tag{29}$$

B. 2PN differences

At second post-Newtonian (2PN) order, the exponential and Schwarzschild metrics differ. Schwarzschild in isotropic coordinates:

$$g_{00}^{\text{Schw}} = -\frac{(1-U)^2}{(1+U)^2} = -(1 - 4U + 8U^2 - 16U^3 + \cdots)\,, \tag{30}$$

$$g_{ij}^{\text{Schw}} = (1+U)^4\,\delta_{ij} = (1 + 4U + 6U^2 + \cdots)\,\delta_{ij}\,. \tag{31}$$

Note that in standard isotropic form, Schwarzschild uses $U_{\text{iso}} = M/(2r_{\text{iso}})$, and the expansion coefficients differ from our $U = r_{\text{s}}/(2r)$ convention.

In the convention $U = GM/(c^2 r)$ (where r is the isotropic radial coordinate), the expansions are:

Coefficient	**Exponential**	**Schwarzschild**
g_{00}: U^1	-2	-2
g_{00}: U^2	$+2$	$+2$
g_{00}: U^3	$-\frac{4}{3}$	$-\frac{8}{3}$
g_{ij}: U^1	$+2$	$+2$
g_{ij}: U^2	$+2$	$+\frac{3}{2}$

The first difference is at $O(U^2)$ in g_{ij} and $O(U^3)$ in g_{00}.

C. Mercury precession

The perihelion precession per orbit for a general static spherically symmetric metric with PPN parameters γ and β is

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)}\left(\frac{2-\beta+2\gamma}{3}\right) + O(U^2)\,, \tag{32}$$

where a is the semi-major axis and e the eccentricity. For both Schwarzschild and the exponential metric, $\gamma = \beta = 1$, giving

$$\Delta\phi_{\text{exp}} = \Delta\phi_{\text{Schw}} = \frac{6\pi GM}{c^2 a(1-e^2)} = 42.98''/\text{cy}\,. \tag{33}$$

At 2PN, the correction to Mercury's precession is [8]:

$$\delta(\Delta\phi) = \Delta\phi_{\text{exp}} - \Delta\phi_{\text{Schw}} \sim \frac{6\pi GM}{c^2 a(1-e^2)} \cdot O\left(\frac{GM}{c^2 a}\right) \approx 2.3 \times 10^{-7}\ \text{arcsec/cy}\,. \tag{34}$$

Current measurement precision is $\sim 3 \times 10^{-3}$ arcsec/cy [9], so the 2PN difference is four orders of magnitude below detectability.

D. Light bending: debunking the "half bending" myth

A persistent misconception holds that the exponential metric "predicts half the observed light bending." This conflates the exponential metric (a solution of GR + phantom scalar) with the Yilmaz field equations (an alternative theory of gravity).

The deflection angle for light in a general static, spherically symmetric metric with PPN parameters γ and β is

$$\delta\theta = \frac{(1+\gamma)}{2} \cdot \frac{4GM}{c^2 b} + O(U^2)\,, \tag{35}$$

where b is the impact parameter. Since $\gamma = 1$ for the exponential metric (proven above):

$$\delta\theta_{\text{exp}} = \frac{4GM}{c^2 b} = \delta\theta_{\text{Schw}}\,. \tag{36}$$

The "half bending" confusion arises from evaluating the bending contribution from g_{00} alone, which gives $2GM/(c^2 b)$. In GR, an equal contribution comes from the spatial curvature (g_{ij} term). The exponential metric has $\gamma = 1$, so both contributions are present and equal:

$$\delta\theta = \underbrace{\frac{2GM}{c^2 b}}_{g_{00}\text{ contribution}} + \underbrace{\frac{2GM}{c^2 b}}_{g_{ij}\text{ contribution}} = \frac{4GM}{c^2 b}\,. \tag{37}$$

The Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ [9] is satisfied exactly by the exponential metric.

III. S3. ECHO DELAY CALCULATION

We compute the gravitational-wave echo delay for the exponential metric in detail.

A. Tortoise coordinate

The tortoise coordinate for the exponential metric $g_{00} = -e^{-r_s/r}$, $g_{rr} = e^{+r_s/r}$ (isotropic coordinates) is

$$r^* = \int \frac{\sqrt{g_{rr}}}{\sqrt{-g_{00}}}\, dr = \int e^{r_s/r}\, dr\,. \tag{38}$$

This integral is finite for all $r > 0$, in sharp contrast to the Schwarzschild tortoise coordinate $r^*_{\text{Schw}} = r + r_s \ln|r/r_s - 1| \to -\infty$ as $r \to r_s$.

B. The echo integral

The round-trip light travel time from the photon sphere at $r_{\text{ph}} = r_s$ (isotropic coordinates) to the wormhole throat at $r_{\text{throat}} = r_s/2$ and back is

$$\Delta t_{\text{echo}} = \frac{2}{c}\int_{r_{\text{throat}}}^{r_{\text{ph}}} e^{r_s/r}\, dr = \frac{2}{c}\int_{r_s/2}^{r_s} e^{r_s/r}\, dr\,. \tag{39}$$

Substituting $u = r_s/r$ (so $r = r_s/u$, $dr = -r_s\, du/u^2$):

$$\Delta t_{\text{echo}} = \frac{2r_s}{c}\int_1^2 \frac{e^u}{u^2}\, du\,. \tag{40}$$

Numerical evaluation.—The integral can be evaluated using the exponential integral function $\text{Ei}(x)$:

$$\int \frac{e^u}{u^2}\, du = -\frac{e^u}{u} + \text{Ei}(u) + C\,. \tag{41}$$

Therefore

$$\begin{aligned}\int_1^2 \frac{e^u}{u^2}\, du &= \left[-\frac{e^u}{u} + \text{Ei}(u)\right]_1^2 \\ &= \left(-\frac{e^2}{2} + \text{Ei}(2)\right) - (-e + \text{Ei}(1)) \\ &= -\frac{e^2}{2} + e + \text{Ei}(2) - \text{Ei}(1)\,. \end{aligned} \tag{42}$$

Using $e^2/2 \approx 3.695$, $e \approx 2.718$, $\text{Ei}(2) \approx 4.954$, $\text{Ei}(1) \approx 1.895$:

$$\int_1^2 \frac{e^u}{u^2}\, du \approx -3.695 + 2.718 + 4.954 - 1.895 = 2.082\,. \tag{43}$$

The echo delay is

$$\boxed{\Delta t_{\text{echo}} \approx \frac{4.16\, r_s}{c}\,.} \tag{44}$$

TABLE II. Echo delays for representative sources. $r_s = 2GM/c^2$.

Source	$M/M_\odot$	r_s (km)	Δt_{echo}
GW150914 remnant	62	183	2.5 ms
GW190521 remnant	150	443	6.1 ms
Intermediate BH	10^3	2,950	41 ms
Sgr A*	4×10^6	1.18×10^7	163 s
M87*	6.5×10^9	1.92×10^{10}	7.4×10^4 s ($\approx$ 21 hr)

C. Numerical estimates for astrophysical sources

D. Comparison with Planck-wall echo model

In the "Planck wall" model of Cardoso et al. [10], the reflecting surface is placed at Planck-length distance from the would-be horizon. The echo delay in Schwarzschild tortoise coordinates is

$$\Delta t_{\text{Planck}} \approx \frac{r_s}{c} \ln\left(\frac{r_s}{\ell_P}\right), \tag{45}$$

where $\ell_P \approx 1.6 \times 10^{-35}$ m is the Planck length. For a $60\, M_\odot$ remnant: $\ln(r_s/\ell_P) \approx \ln(10^{38}) \approx 88$, giving $\Delta t_{\text{Planck}} \approx 88\, r_s/c \approx 50$ ms.

The exponential-metric echo is approximately 20× shorter than the Planck-wall echo:

$$\frac{\Delta t_{\text{echo}}}{\Delta t_{\text{Planck}}} \approx \frac{4.2}{\ln(r_s/\ell_P)} \sim \frac{1}{20}\,. \tag{46}$$

This difference is observationally significant: current echo searches [11] use templates with $\Delta t \sim 200$ ms, appropriate for Planck-wall models but potentially missing the shorter exponential-metric signal.

IV. S4. PETZ SATURATION ANALYSIS

A. Why exact saturation is impossible for $\Sigma > 0$

Theorem 1 (No exact saturation for $\Sigma > 0$)**.** *Let $\mathcal{N}$ be a CPTP map and σ a faithful state. The JRSWW bound [12]*

$$F(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}} \circ \mathcal{N}(\rho))^2 \geq \exp(-\Delta D) \tag{47}$$

is saturated (equality holds for all ρ) if and only if $\Delta D = 0$, i.e., $\mathcal{N}$ is sufficient for σ.

Proof. The JRSWW proof proceeds via the Golden–Thompson inequality [13]:

$$\mathrm{Tr}[e^{A+B}] \leq \mathrm{Tr}[e^A e^B]\,, \tag{48}$$

with equality if and only if $[A, B] = 0$. In the application to the Petz recovery bound, A and B involve $\log\sigma$ and $\log\mathcal{N}(\sigma)$; these commute if and only if the channel is sufficient for σ, which by Petz's theorem occurs precisely when $\Delta D = 0$ [14].

More precisely, the chain of inequalities in the JRSWW proof is:

$$\begin{aligned} &F(\rho, \tilde{\mathcal{R}} \circ \mathcal{N}(\rho))^2 \\ &\quad \geq \exp(-D_M(\rho\|\sigma)) \cdot \exp(D_M(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))) \\ &\quad = \exp(-\Delta D_M) \geq \exp(-\Delta D)\,, \end{aligned} \tag{49}$$

where D_M is the measured relative entropy and $\Delta D_M \leq \Delta D$ with equality for commuting states. The first inequality uses the Golden–Thompson inequality and becomes an equality only when $[\log\sigma, \mathcal{N}^\dagger(\log\mathcal{N}(\sigma))] = 0$, which for faithful σ is equivalent to $\Delta D = 0$. □

Remark. *This theorem means that for any gravitational channel with $\Sigma_{\mathrm{grav}} > 0$, the Petz recovery fidelity strictly exceeds $\exp(-\Sigma_{\mathrm{grav}}/2)$. The exponential metric, defined by promoting $F = \exp(-\Sigma_{\mathrm{grav}}/2)$ to an identity, should be understood as a* lower bound geometry*—the metric that would result if recovery were as poor as the JRSWW bound allows. Nature may do better, and the actual metric may lie between the exponential and Schwarzschild predictions.*

B. Gap calculation

For a weak gravitational field ($\Sigma_{\mathrm{grav}} = r_{\mathrm{s}}/r \ll 1$), we can estimate the gap between the actual Petz recovery fidelity and the JRSWW bound.

Consider the amplitude damping channel as a model for gravitational loss. The amplitude damping channel with parameter $\gamma = 1 - \eta$ (where $\eta = e^{-r_{\mathrm{s}}/r} \approx 1 - r_{\mathrm{s}}/r$ for weak fields) has Kraus operators

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\eta} \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{1-\eta} \\ 0 & 0 \end{pmatrix}. \tag{50}$$

For the maximally mixed reference state $\sigma = I/2$, the Petz recovery map is

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) = \frac{1}{2}\sum_k K_k^\dagger \left(\frac{I}{2}\right)^{-1/2} \rho \left(\frac{I}{2}\right)^{-1/2} K_k = \sum_k K_k^\dagger \rho\, K_k\,. \tag{51}$$

The fidelity F_{Petz} for the input $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ can be computed explicitly. Writing $\epsilon = 1 - \eta = r_{\text{s}}/r$:

$$F_{\text{Petz}}^2 = 1 - \frac{1}{4}\epsilon^2 + O(\epsilon^3)\,, \tag{52}$$

$$\exp(-\Sigma_{\text{grav}}) = \exp(-\epsilon) = 1 - \epsilon + \frac{1}{2}\epsilon^2 + O(\epsilon^3)\,. \tag{53}$$

The gap is

$$\boxed{\Delta \equiv F_{\text{Petz}}^2 - \exp(-\Sigma_{\text{grav}}) = \epsilon - \frac{3}{4}\epsilon^2 + O(\epsilon^3) = \frac{r_{\text{s}}}{r} - \frac{3}{4}\left(\frac{r_{\text{s}}}{r}\right)^2 + \cdots\,.} \tag{54}$$

The leading gap is $O(r_{\text{s}}/r)$, not $O((r_{\text{s}}/r)^2)$. This means the Petz recovery does significantly better than the JRSWW bound in weak fields. The relative gap $\Delta/\exp(-\Sigma_{\text{grav}}) \approx r_{\text{s}}/r$ is small, confirming approximate saturation.

C. Schwarzschild fidelity vs. exponential fidelity

For the Schwarzschild metric, $F_{\text{bound}} = \sqrt{1 - r_{\text{s}}/r}$:

$$F_{\text{Schw}} = \sqrt{1 - r_{\text{s}}/r} = 1 - \frac{1}{2}\frac{r_{\text{s}}}{r} - \frac{1}{8}\left(\frac{r_{\text{s}}}{r}\right)^2 + \cdots \tag{55}$$

For the exponential metric, $F_{\text{bound}} = e^{-r_{\text{s}}/(2r)}$:

$$F_{\text{exp}} = e^{-r_{\text{s}}/(2r)} = 1 - \frac{1}{2}\frac{r_{\text{s}}}{r} + \frac{1}{8}\left(\frac{r_{\text{s}}}{r}\right)^2 + \cdots \tag{56}$$

The difference is

$$F_{\text{Schw}} - F_{\text{exp}} = -\frac{1}{4}\left(\frac{r_{\text{s}}}{r}\right)^2 + O\left(\left(\frac{r_{\text{s}}}{r}\right)^3\right). \tag{57}$$

At the neutron star surface ($r_{\text{s}}/r \approx 0.4$): $|F_{\text{Schw}} - F_{\text{exp}}| \approx 0.04$, a 5% difference. At $r_{\text{s}}/r = 1$ (horizon): $F_{\text{Schw}} = 0$ but $F_{\text{exp}} = e^{-1/2} \approx 0.607$.

V. S5. VAIDYA CALCULATION: $\tau = 1$ AT THE APPARENT HORIZON

We provide the complete derivation of the Kodama vector and its connection to τ in the Vaidya spacetime, complementing the abbreviated treatment in the existing supplement.

A. Vaidya metric setup

The ingoing Vaidya metric in advanced Eddington–Finkelstein coordinates (v, r, θ, ϕ) is

$$ds^2 = -f(v,r)\,dv^2 + 2\,dv\,dr + r^2\,d\Omega^2\,, \tag{58}$$

where $f(v,r) = 1 - 2M(v)/r$ and $M(v)$ is the Bondi mass.

The areal radius is $R = r$. The 2D orbit-space metric on (v, r) is

$$h_{AB}\,dx^A\,dx^B = -f\,dv^2 + 2\,dv\,dr\,. \tag{59}$$

The Kodama vector is $K_A = \epsilon_A{}^B\,\partial_B R$ with $\epsilon_{vr} = 1$, giving (as derived in the companion supplement):

$$K^a\partial_a = -\partial_v\,, \qquad |K| = \sqrt{f} = \sqrt{1 - 2M(v)/r}\,. \tag{60}$$

B. $g_{00} \to 0$ at the MOTS

The marginally outer trapped surface (MOTS) is defined by the vanishing of the expansion of outgoing null geodesics. For the Vaidya metric (58), the outgoing null expansion is

$$\theta_+ = \frac{2}{r}\left(1 - \frac{2M(v)}{r}\right)^{1/2} \cdot \frac{1}{r}\left(r - 2M(v)\right)\,. \tag{61}$$

More precisely, the expansion of the outgoing null normal $\ell^a = \partial_v + \frac{f}{2}\partial_r$ is

$$\theta_+ = \frac{f}{r} = \frac{1}{r}\left(1 - \frac{2M(v)}{r}\right). \tag{62}$$

This vanishes when $r = 2M(v)$, i.e., at the MOTS (apparent horizon).

At the MOTS:

$$f\big|_{r=2M(v)} = 1 - \frac{2M(v)}{2M(v)} = 0\,, \tag{63}$$

so $|K| = 0$ and

$$\tau_K = 1 - |K| = 1 - 0 = 1\,. \tag{64}$$

Complete information loss at the apparent horizon.

C. Connection to quantum extremal surface

The quantum extremal surface (QES) [15] extremizes the generalized entropy $S_{\text{gen}} = A/(4G\hbar) + S_{\text{bulk}}$. In the semiclassical limit, the QES coincides with the MOTS. Including quantum corrections:

$$r_{\text{QES}} = 2M(v) + \frac{4G\hbar}{r} \cdot \frac{\partial S_{\text{bulk}}}{\partial r}\bigg|_{r=2M(v)} + O(\hbar^2)\,. \tag{65}$$

The quantum-corrected τ at the QES is

$$\tau_K(r_{\text{QES}}) = 1 - O(\hbar/M^2) < 1\,, \tag{66}$$

so the quantum correction makes the information loss slightly incomplete—consistent with the expectation that quantum gravity effects prevent complete information loss.

VI. S6. PIKOVSKI CHANNEL ANALYSIS

We provide a complete worked example of the Pikovski gravitational dephasing channel, including the Petz recovery map, fidelity calculations, and a detailed argument for why it does *not* yield the gravitational channel $\mathcal{N}_{\text{grav}}$.

A. The Pikovski dephasing channel

Pikovski et al. [16] showed that gravitational time dilation universally dephases composite quantum systems. A particle in a superposition of heights Δh in a gravitational field g experiences a phase difference

$$\Delta\phi = \frac{mc^2\, g\, \Delta h\, t}{\hbar c^2} = \frac{\Delta E\, \Delta\Phi\, t}{\hbar c^2}\,, \tag{67}$$

where ΔE is the internal energy spread and $\Delta\Phi = g\,\Delta h$ is the gravitational potential difference.

For a system with internal energy distribution $\{E_n, p_n\}$, the coherence between height states $|h_1\rangle$ and $|h_2\rangle$ decays as

$$\rho_{12}(t) = \rho_{12}(0) \cdot \chi(\Delta t_{\text{grav}})\,, \tag{68}$$

where $\chi(\tau) = \sum_n p_n\, e^{-iE_n\tau/\hbar}$ is the characteristic function of the internal energy distribution and $\Delta t_{\rm grav} = g\,\Delta h\, t/c^2$ is the proper-time difference.

Coherence parameter.—The dephasing channel has a single coherence parameter

$$p = |\chi(\Delta t_{\rm grav})|\,. \tag{69}$$

Two-level system at zero temperature.—For a two-level internal system with gap ω_0:

$$p = |\cos(\omega_0\,\Delta t_{\rm grav}/2)|\,. \tag{70}$$

Thermal state.—For a thermal internal state at temperature T:

$$p = \left|\frac{Z(\beta - i\Delta t_{\rm grav}/\hbar)}{Z(\beta)}\right|, \tag{71}$$

where $Z(\beta) = \mathrm{Tr}[e^{-\beta H}]$ is the partition function.

B. Kraus operators

The qubit dephasing channel with parameter p has Kraus operators

$$K_0 = \sqrt{\frac{1+p}{2}}\,I\,, \qquad K_1 = \sqrt{\frac{1-p}{2}}\,\sigma_z\,, \tag{72}$$

satisfying $K_0^\dagger K_0 + K_1^\dagger K_1 = I$. The channel acts as

$$\mathcal{N}_p(\rho) = p\,\rho + (1-p)\,\frac{\sigma_z\rho\sigma_z + \rho}{2} = \begin{pmatrix} \rho_{00} & p\,\rho_{01} \\ p\,\rho_{10} & \rho_{11} \end{pmatrix}. \tag{73}$$

C. Petz recovery map for dephasing

For the maximally mixed reference $\sigma = I/2$:

Step 1: $\mathcal{N}_p(\sigma) = \sigma = I/2$.

Step 2: $\mathcal{N}_p(\sigma)^{-1/2} = \sqrt{2}\,I$.

Step 3: The Petz map is

$$\begin{aligned}\mathcal{R}_{\sigma,\mathcal{N}_p}(\rho) &= \sigma^{1/2}\,\mathcal{N}_p^\dagger\big(\mathcal{N}_p(\sigma)^{-1/2}\,\rho\,\mathcal{N}_p(\sigma)^{-1/2}\big)\,\sigma^{1/2} \\ &= \frac{1}{2}\,\mathcal{N}_p^\dagger(2\rho) = \mathcal{N}_p(\rho)\,.\end{aligned} \tag{74}$$

The key observation: for diagonal σ, the Petz recovery map for the dephasing channel *equals the channel itself.* The Petz map applies dephasing *again*, making things worse rather than better.

D. Fidelity calculations

For the input state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$:

After the channel: $\mathcal{N}_p(|+\rangle\langle+|) = \frac{1}{2}\left(\begin{smallmatrix} 1 & p \\ p & 1 \end{smallmatrix}\right)$.

After Petz recovery:

$$\mathcal{R} \circ \mathcal{N}_p(|+\rangle\langle+|) = \mathcal{N}_p^2(|+\rangle\langle+|) = \frac{1}{2}\begin{pmatrix} 1 & p^2 \\ p^2 & 1 \end{pmatrix}. \tag{75}$$

Petz fidelity:

$$F_{\text{Petz}} = \langle+|\mathcal{R} \circ \mathcal{N}_p(|+\rangle\langle+|)|+\rangle = \frac{1+p^2}{2}. \tag{76}$$

Optimal fidelity (identity map—do nothing):

$$F_{\text{opt}} = \langle+|\mathcal{N}_p(|+\rangle\langle+|)|+\rangle = \frac{1+p}{2}. \tag{77}$$

The Petz map is *worse* than doing nothing: $F_{\text{Petz}} < F_{\text{opt}}$ for $0 < p < 1$. This is because the Petz map is designed for the reference state σ, not for the input state ρ; for dephasing with diagonal σ, the Petz map coincides with the noise itself.

The JRSWW bound:

$$\Sigma_p = D(|+\rangle\langle+|\|\sigma) - D(\mathcal{N}_p(|+\rangle\langle+|)\|\mathcal{N}_p(\sigma)) = \ln 2 - H_{\text{bin}}\left(\frac{1+p}{2}\right), \tag{78}$$

where H_{bin} is the binary entropy. For p close to 1 ($\epsilon = 1 - p \ll 1$): $\Sigma_p \approx \epsilon^2/(2\ln 2)$.

E. Why Pikovski is NOT the gravitational channel

There are three fundamental mismatches between the Pikovski dephasing channel and the gravitational entropy production $\Sigma_{\text{grav}} = r_{\text{s}}/r$:

Mismatch 1: Probe dependence.—The Pikovski entropy production

$$\Sigma_{\text{Pik}} \propto (\Delta E)^2 (\Delta\Phi)^2 t^2 \tag{79}$$

depends on the internal energy spread ΔE, the mass m, and the observation time t. In contrast, $\Sigma_{\text{grav}} = r_{\text{s}}/r$ is *universal*—independent of the probe.

Mismatch 2: Divergent supremum.—Taking $\Delta E \to \infty$ gives $\Sigma_{\text{Pik}} \to \infty$ at any fixed $\Delta\Phi$ and t. No finite supremum exists. In contrast, Σ_{grav} is finite for all $r > 0$.

Mismatch 3: Basis.—The Pikovski channel dephases in the *position* basis of center-of-mass motion. The gravitational channel should dephase (or lose information) in the *energy/frequency* basis, corresponding to the redshift of photon frequencies.

What the Pikovski channel IS.—The Pikovski dephasing rate per unit frequency per unit time is $\Gamma = g\,\Delta h/c^2 = \Delta\Phi/c^2 \sim \Sigma_{\rm grav}/2$. This universal rate is correctly predicted by $\Sigma_{\rm grav}$, but the *accumulated* dephasing depends on the specific probe. The Pikovski channel is a *consequence* of gravitational information loss, not the gravitational channel itself.

F. Numerical estimates

For a molecule of mass $m = 10^4$ amu with internal energy spread $\Delta E = 1$ eV at height difference $\Delta h = 1$ m on Earth's surface ($g \approx 9.8$ m/s^2):

$$\Delta t_{\rm grav} = \frac{g\,\Delta h}{c^2}\,t \approx 1.1\times 10^{-16}\,t\ (\text{seconds})\,, \tag{80}$$

$$p = \cos(\omega_0\,\Delta t_{\rm grav}/2) \approx 1 - \frac{(\Delta E)^2(\Delta t_{\rm grav})^2}{8\hbar^2}\,. \tag{81}$$

For $t = 1$ s: $\Delta t_{\rm grav} \approx 10^{-16}$ s, $\omega_0 \approx 1.5\times 10^{15}$ rad/s (for $\Delta E = 1$ eV), $\omega_0\,\Delta t_{\rm grav}/2 \approx 7.6\times 10^{-2}$ rad, $p \approx 0.997$, $\tau_{\rm Pik} \approx 0.003$.

The corresponding gravitational τ at the Earth's surface is $\tau_{\rm grav} = 1 - \sqrt{1 - r_{{\rm s},\oplus}/R_\oplus} \approx 7\times 10^{-10}$—six orders of magnitude smaller. This confirms that the Pikovski dephasing accumulates much faster than the static gravitational τ for a probe-dependent reason (the internal energy spread amplifies the effect).

VII. S7. RAYCHAUDHURI DECOMPOSITION

A. Raychaudhuri equation and entropy production

For a congruence of timelike geodesics with tangent vector u^a, the Raychaudhuri equation reads [18, 19]

$$\frac{d\theta}{d\tau} = -\frac{\theta^2}{3} - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}\,u^a u^b\,, \tag{82}$$

where θ is the expansion scalar, σ_{ab} the shear tensor, and ω_{ab} the vorticity tensor.

Interpreting each term as a contribution to the rate of change of gravitational entropy production [17]:

$$\frac{d\Sigma_{\rm grav}}{ds} = \underbrace{\frac{\theta^2}{3}}_{\text{expansion}} + \underbrace{\sigma_{ab}\sigma^{ab}}_{\text{shear}} - \underbrace{\omega_{ab}\omega^{ab}}_{\text{vorticity}} + \underbrace{R_{ab}\,u^a u^b}_{\text{curvature}} . \tag{83}$$

B. Static vacuum Schwarzschild: $\Sigma_{\rm grav,Raych} = 0$

For static observers in the Schwarzschild vacuum:

- $\theta = 0$ (static congruence, no expansion);
- $\sigma_{ab} = 0$ (spherical symmetry, no shear);
- $\omega_{ab} = 0$ (hypersurface-orthogonal, no vorticity);
- $R_{ab} = 0$ (vacuum solution).

Therefore $d\Sigma_{\rm grav}/ds = 0$: the Raychaudhuri decomposition gives *zero* entropy production rate.

This does *not* contradict $\Sigma_{\rm grav} = r_{\rm s}/r$; rather, it shows that $\Sigma_{\rm grav}$ from the modular flow route is an *integrated* quantity (the total QRE decrease from r to ∞), while the Raychaudhuri decomposition gives the *local rate*. The two are complementary, not contradictory:

$$\Sigma_{\rm grav} = \int_r^\infty \frac{d\Sigma_{\rm grav}}{ds}\, ds + \text{boundary term} . \tag{84}$$

The boundary term (from the Tolman redshift at infinity) provides $\Sigma_{\rm grav} = -\ln(-g_{00})$; the local Raychaudhuri rate vanishes in vacuum.

C. NEC violation implies $\tau < 0$

Proposition 2 (One-directional NEC–τ connection)**.** *If the null energy condition (NEC) is violated along a null geodesic congruence, i.e., $R_{ab}\,k^a k^b < 0$ for some null vector k^a, then there exist configurations where $d\Sigma_{\rm grav}/d\lambda < 0$, implying τ can decrease along the congruence.*

Proof. For a null geodesic congruence with tangent k^a, the null Raychaudhuri equation (with vorticity vanishing by the Frobenius theorem for hypersurface-orthogonal null congruences) is

$$\frac{d\theta}{d\lambda} = -\frac{\theta^2}{2} - \hat{\sigma}_{ab}\hat{\sigma}^{ab} - R_{ab}\,k^a k^b . \tag{85}$$

If $R_{ab}\,k^a k^b < 0$ (NEC violation), the last term is positive, potentially making $d\theta/d\lambda > 0$ (defocusing).

In the entropy production interpretation:

$$\frac{d\Sigma_{\rm grav}}{d\lambda} = \frac{\theta^2}{2} + \hat{\sigma}_{ab}\hat{\sigma}^{ab} + R_{ab}\,k^a k^b\,. \tag{86}$$

If NEC is violated strongly enough that $|R_{ab}\,k^a k^b| > \theta^2/2 + \hat{\sigma}^2$, then $d\Sigma_{\rm grav}/d\lambda < 0$: entropy production decreases, and τ can decrease. This corresponds to information flowing *back* from the environment to the system—a form of non-Markovian backflow. □

Remark. *The converse does not hold: $d\Sigma_{\rm grav}/d\lambda < 0$ does not require NEC violation (it could result from decreasing shear or expansion). The NEC–τ connection is one-directional: NEC violation $\Rightarrow$ possible τ decrease, but not conversely.*

D. Kerr vorticity contribution

For the Kerr metric with spin parameter a, the congruence of static observers at Boyer–Lindquist radius r has nonzero vorticity:

$$\omega^2 = \omega_{ab}\omega^{ab} = \frac{2M^2a^2r^2}{\Sigma^4}\,, \tag{87}$$

where $\Sigma^2 = r^2 + a^2\cos^2\theta$. The vorticity term $-\omega^2$ in the Raychaudhuri equation acts to *decrease* $d\Sigma_{\rm grav}/ds$:

$$\left.\frac{d\Sigma_{\rm grav}}{ds}\right|_{\rm Kerr} = R_{ab}\,u^a u^b - \omega^2 < R_{ab}\,u^a u^b\,. \tag{88}$$

Frame dragging (vorticity) reduces the effective entropy production rate, consistent with the physical intuition that angular momentum protects information. In the extremal Kerr limit ($a \to M$), the vorticity contribution becomes significant near the (would-be) horizon, potentially allowing $\tau < 1$ at $r = r_+$.

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